

MAHINDRA IRVAN SAPUTRA

Batang, Centra Java | +62 888 690 140 | mahindra.irvan538@gmail.com
linkedin.com/in/mahindra-irvan-saputra-7925941aa/ | https://github.com/zippo538

Profile

Computer Engineering graduate and Software Engineer with strong experience in building end-to-end web applications using C#, .NET Core, Node.js, and PostgreSQL. Experienced in applying Clean Architecture, Design Patterns, and efficient database designs. Formulatrix Bootcamp Batch 20 Alumni with a proven track record in developing scalable backend systems, complex business logic, and web integrations.

TECHNICAL PROJECTS

Bootcamp Formulatrix Batch 20

Aug 2026 – Sep 2026

Salatiga, Jawa Tengah

Lab Consumable Expiry Tracker (FEFO System)

- Built a full-stack inventory management system using React (TypeScript, Tailwind CSS) on the frontend and .NET Web API (C#) on the backend..
- Architected scalable backend logic adhering to Clean Architecture, Dependency Injection, Repository Pattern, FluentValidation, and Entity Framework Core.
- Developed an automated First-Expired, First-Out (FEFO) allocation algorithm integrated with PostgreSQL to minimize chemical waste and optimize usage.
- Implemented Unit of Work and Optimistic Concurrency Control (RowVersion) to resolve database race conditions and ensure data integrity under concurrent access.
- Applied Specification-Driven Development (SDD) to structure Product Requirement Documents (PRDs) for AI-guided code generation.
- Secured application routes and endpoints using end-to-end JWT-based authentication and client-side route guards
- Ensured end-to-end type safety between client and API layers via strongly-typed React UI components, custom hooks, and interface models.

Build Chess Game

- Designed a dual-interface C# chess application featuring both a Console CLI (Spectre.Console) and a React Web Client.
- Implemented Domain-Driven Design (DDD) principles and microservice-oriented patterns with integrated logging and error handling.
- Integrated NUnit testing suites and modular game engine logic to validate chess moves, state management, and board conditions

Bootcamp Machine Learning / Artificial Intelligence Dibimbing

Jun 2025 – Des 2025

Online

NLP Chatbot with sentiment analysis

- Developed an NLP chatbot using the Haystack LLM framework to reject LLM answer negative or bad word government.
- Implemented a production-ready Machine Learning workflow featuring data preprocessing, model training with hyperparameter tuning, experiment tracking, and automated deployment.
- Integrated MLflow into an MLOps pipeline for experiment tracking, metric logging, and model versioning
- Engineered modular and scalable system architectures adhering to clean code principles for production readiness.

House Price Prediction Platform (MLOps)

- Built an end-to-end house price prediction web application (Streamlit) utilizing machine learning algorithms and hyperparameter tuning.
- Containerized and deployed the solution using Docker and FastAPI for portable, scalable, and async model serving.
- Integrated MLflow for experiment tracking, metric logging, and model versioning as part of an MLOps pipeline.

Bootcamp TSA KOMINFO x Rakamin

Aug 2023 – Des 2023

Online

- Developed the information system Peer-to-Peer (P2P) lending credit system using Laravel and PHP Filament to meet lender and borrower. The feature payment gateway, verification approve borrower by admin, Loan Application, and paying debt with interest 5% each month.

Work Experience

Bimbel Citra Bhakti Pertiwi

Oct 2023 – Jul 2026

Batang, Jawa Tengah

Administration Staff

- Streamlined institutional data workflows and administrative documentation to support daily academic operations.
- Instructed 30+ students in General Intelligence (TIU) and psychological test preparation, mentoring them in analytical problem-solving strategies.
- Managed candidate tracking data, contributing to 31 students successfully passing civil service selections over 3 years.

EDUCATION

Institut Widya Pratama

Nov 2021 – Nov 2025

Pekalongan, Jawa Tengah

Bachelor Degree in Computer Engineering GPA: 3.78 / 4.00

Relevant Focus Areas: Artificial Intelligence, Machine Learning, System Architecture, Design Patterns, Data Structures Algorithms

Organizational Experience

Dewan Perwakilan Mahasiswa

Des 2023 – Des 2024

Pekalongan, Jawa Tengah

- Led accountability evaluations for 8 executive student bodies and facilitated strategic budget allocations.

Badan Eksekutif Mahasiswa

Feb 2022 – Aug 2022

Pekalongan, Jawa Tengah

- Facilitated student advocacy forums and coordinated campus-wide engagement events.

SKILLS

Technical Skills:

- **Languages:** C#, TypeScript, JavaScript, Python, PHP, Java
- **Frameworks & Libraries:** .NET Core / ASP.NET, React, Node.js, FastAPI, Laravel
- **Database :** Mysql, PostgreSQL, SQLite
- **DevOps & Tools:** Docker, Git, AWS (Cloud Basics), RESTful APIs, Postman

CERTIFICATIONS

- Bootcamp Formulatrix C# Batch 20 – Formulatrix (2026)
- Bootcamp Machine Learning / Artificial Intelligence Batch 7 – Dibimbing (2025)
- Cloud Practitioner Essentials (AWS Cloud Basics) – Dicoding Indonesia (2023)
- Back-End Application Development for Beginners – Dicoding Indonesia (2023)